International and Monetary Economics

Course Code: 30054

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Lecture 10

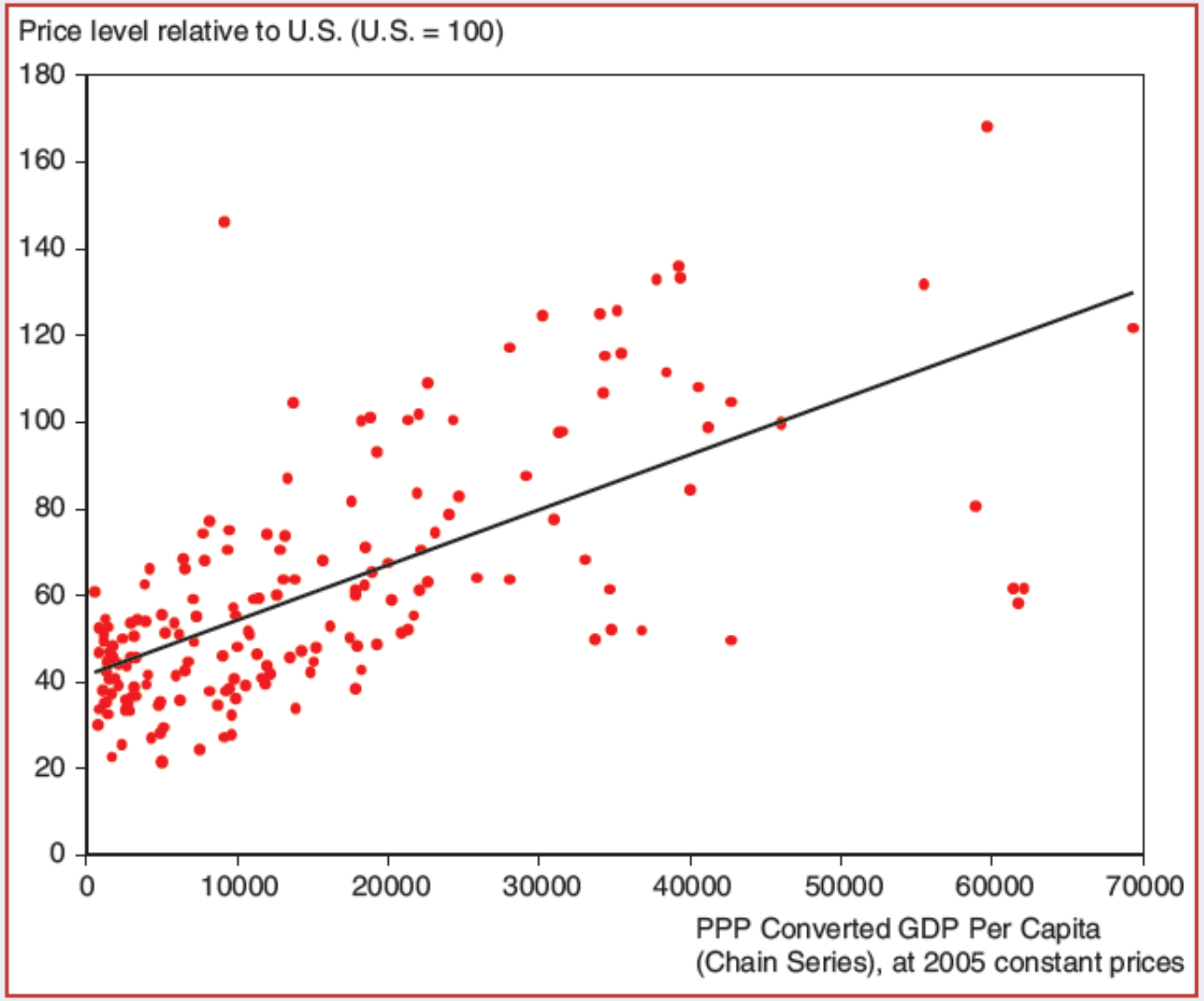
Roadmap

1. Price Levels and Income
2. Real exchange rate
   - Real interest rate parity
3. A General Theory of FX in the Long Run
4. From fixed to variable output
   - Consumption
   - Net exports
     - How do changes in FX affect *NX*?

       [PCP perspective]

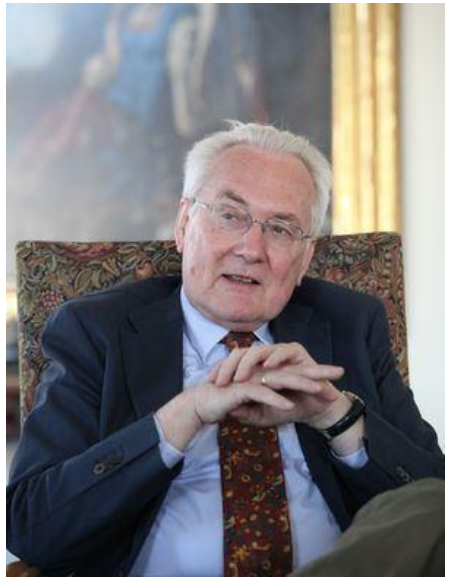
Price levels and Income

A Striking Regularity

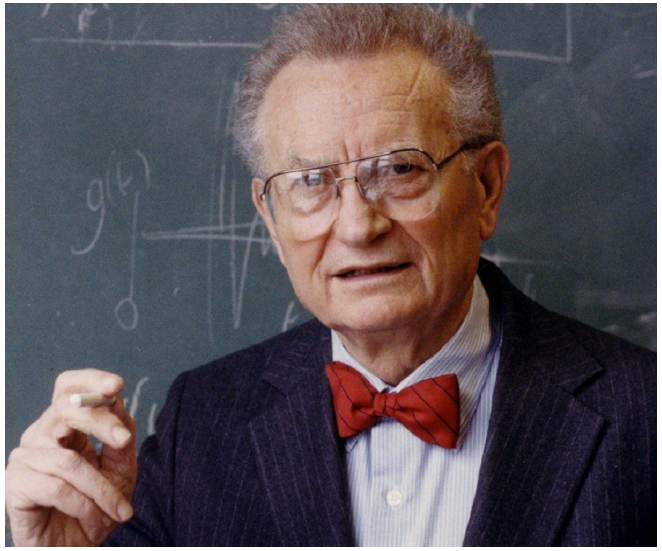


Price levels are lower in poor countries. Why?

The Balassa-Samuelson Effect



Béla Balassa (1928-1991)



Paul Samuelson (1915-2009)

- Nobel Prize 1970 (he "*has done more than any other contemporary economist to raise the level of scientific analysis in economic theory*")

The Balassa-Samuelson Effect

- **Assumptions**
  - Poor countries are less productive in producing tradable goods than rich countries (cars, computers, etc.)
  - Productivity differences are negligible in the non-traded sector (services: haircuts, nannies, etc.)
- **Intuition behind the Balassa-Samuelson effect**
  - If traded goods satisfy the Law of One Price (LOP), then lower traded goods productivity implies lower wages (in terms of traded goods), which implies lower prices of non-traded goods (in terms of traded goods), hence, the overall price level is lower (relative to a foreign country)

The Balassa-Samuelson Effect

(notes posted online)

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The Balassa-Samuelson Effect (Lecture 9)

Notation.

- i - a country index;
- j - a sector index which can be either N (non-traded) or T (traded);
- Y_i^j - output of sector j in country i;
- A_i^j - labor productivity of sector j in country i;
- L_i^j - employment of sector j in country i;
- P_i^j - nominal price of output of sector j in country i;
- P_i - aggregate price index in country i;
- W_i - nominal wage in country i;

Technology. Firms produce output according to the following linear production function

$$Y_i^j = A_i^j L_i^j.$$

Firm's problem. A (representative) firm in sector j and country i maximizes its static profits

$$\max_{L_i^j} \; P_i^j A_i^j L_i^j - W_i L_i^j.$$

Equilibrium. When firms optimize and the markets clear, it must be that marginal revenue equals the marginal cost. Formally,

$$P_i^T A_i^T = P_i^N A_i^N = W_i.$$

1

Price index in country i. Assume that the price level in country i is weighted (geometric) average of traded and non-traded goods prices:

$$P_i = \left(P_i^T\right)^{\alpha} \left(P_i^N\right)^{1-\alpha}.$$

The ratio of price indexes in two countries i and k

$$\frac{P_i}{P_k} = \frac{\left(P_i^T\right)^{\alpha} \left(P_i^N\right)^{1-\alpha}}{\left(P_k^T\right)^{\alpha} \left(P_k^N\right)^{1-\alpha}} = \frac{P_i^T \left(P_i^N / P_i^T\right)^{1-\alpha}}{P_k^T \left(P_k^N / P_k^T\right)^{1-\alpha}} = \frac{P_i^T \left(A_i^T / A_i^N\right)^{1-\alpha}}{P_k^T \left(A_k^T / A_k^N\right)^{1-\alpha}} = E_{i/k} \frac{\left(A_i^T / A_i^N\right)^{1-\alpha}}{\left(A_k^T / A_k^N\right)^{1-\alpha}}.$$

The last equality assumed that the law of one price holds for traded goods, i.e., $P_i^T / P_k^T = E_{i/k}$. After dividing the last equation by $E_{i/k}$, we obtain

$$\frac{P_i}{P_k E_{i/k}} = \frac{\left(A_i^T / A_i^N\right)^{1-\alpha}}{\left(A_k^T / A_k^N\right)^{1-\alpha}}.$$

Balassa and Samuelson assumed that productivity of non-traded sectors are similar across countries, that is, $A_i^N = A^N$ for all i. As a result,

$$\frac{P_i}{P_k E_{i/k}} = \left(\frac{A_i^T}{A_k^T}\right)^{1-\alpha}.$$

The last equation is the main result. It states that the price level in country i relative to the price level in country k, after converting it into currency of country i, is positively related to the ratio of productivities in traded sectors.

2

The Balassa-Samuelson Effect

- Price index in country *i*

$$P_i = \left(P_i^T\right)^{\alpha} \left(P_i^N\right)^{1-\alpha}$$

- Ratio of price indexes in countries *i* and *k (+algebra)*

$$\frac{P_i}{P_k} = \frac{\left(P_i^T\right)^{\alpha} \left(P_i^N\right)^{1-\alpha}}{\left(P_k^T\right)^{\alpha} \left(P_k^N\right)^{1-\alpha}} = \frac{P_i^T \left(P_i^N / P_i^T\right)^{1-\alpha}}{P_k^T \left(P_k^N / P_k^T\right)^{1-\alpha}}$$

The Balassa-Samuelson Effect

- Technology

$$Y_i^j = A_i^j L_i^j$$

Sector index: T (tradable) or NT (non-tradable)

Country index: e.g., Italy

- Firms

$$\max_{L_i^j} \; P_i^j A_i^j L_i^j - W_i L_i^j$$

Wage does not depend on sector j because workers can switch sectors

- Equilibrium

$$P_i^T A_i^T = P_i^N A_i^N = W_i \longrightarrow \frac{P_i^N}{P_i^T} = \frac{A_i^T}{A_i^N}$$

The Balassa-Samuelson Effect

- The LOP is assumed to hold for traded goods

$$P_i^T / P_k^T = E_{i/k}$$

- Price indexes again

$$\frac{P_i}{P_k} = \frac{(P_i^T)^{\alpha}(P_i^N)^{1-\alpha}}{(P_k^T)^{\alpha}(P_k^N)^{1-\alpha}} = \frac{P_i^T(P_i^N/P_i^T)^{1-\alpha}}{P_k^T(P_k^N/P_k^T)^{1-\alpha}} = \boxed{\frac{P_i^T(A_i^T/A_i^N)^{1-\alpha}}{P_k^T(A_k^T/A_k^N)^{1-\alpha}} = E_{i/k}\frac{(A_i^T/A_i^N)^{1-\alpha}}{(A_k^T/A_k^N)^{1-\alpha}}}$$

$$\frac{P_i}{P_k E_{i/k}} = \frac{(A_i^T/A_i^N)^{1-\alpha}}{(A_k^T/A_k^N)^{1-\alpha}} \xrightarrow{\boxed{A_i^N = A^N}} \frac{P_i}{P_k E_{i/k}} = \left(\frac{A_i^T}{A_k^T}\right)^{1-\alpha}$$

The Bhagwati-Kravis-Lipsey Effect

- Rich countries have higher level of capital relative to labor compared to poorer countries

$$\left(\frac{K}{L}\right)^{rich} > \left(\frac{K}{L}\right)^{poor} \Rightarrow$$

- Higher labor productivity (marginal product of labor to be precise)

$$MPL^{rich} > MPL^{poor} \Rightarrow$$

- Higher wages: $w^{rich} > w^{poor} \Rightarrow$
- Because non-tradable goods are more labor intensive, prices of non-tradable goods are higher in richer countries $\Rightarrow$
- Overall price level is higher in richer countries

Real Exchange Rate

Real Exchange Rate

- Nominal exchange rate – the relative price of currencies
- Real exchange rate – the relative price of two baskets of goods

$$q_{\$/€} = \frac{E_{\$/€} \cdot P_E}{P_{US}}$$

  – The US is the home country, and Europe is the foreign country
  – P_{US} is the price of a *typical* US basket of goods
  – P_E is the price of a *typical* Euro basket of goods

Real Exchange Rate

Example

$$q_{\$/€} = \frac{E_{\$/€} \cdot P_E}{P_{US}} = \frac{1.13\,\frac{\$}{€} \cdot 100\,\frac{€}{basket\ of\ European\ goods}}{50\,\frac{\$}{basket\ of\ US\ goods}}$$

$$= 2.26\,\frac{basket\ of\ US\ goods}{basket\ of\ European\ goods}$$

- If $q_{\$/€}$ goes up, we call it a **real** depreciation of the dollar against the Euro

Real Interest Rate Parity

Interest Rate Differences

- Real exchange rate

$$q = \frac{EP^*}{P}$$

- In rate of (expected) change:

$$\frac{q^e - q}{q} \approx \frac{E^e - E}{E} - (\pi^e - \pi^{*e})$$

- Use the UIP, i.e., $R - R^* = \frac{E^e - E}{E}$, to obtain

$$R - R^* = \frac{q^e - q}{q} + (\pi^e - \pi^{*e})$$

Real Interest Rate Parity

- The last equation from the previous slide

$$R - R^* = \frac{q^e - q}{q} + (\pi^e - \pi^{*e})$$

- Real interest rate

$$r = R - \pi^e$$

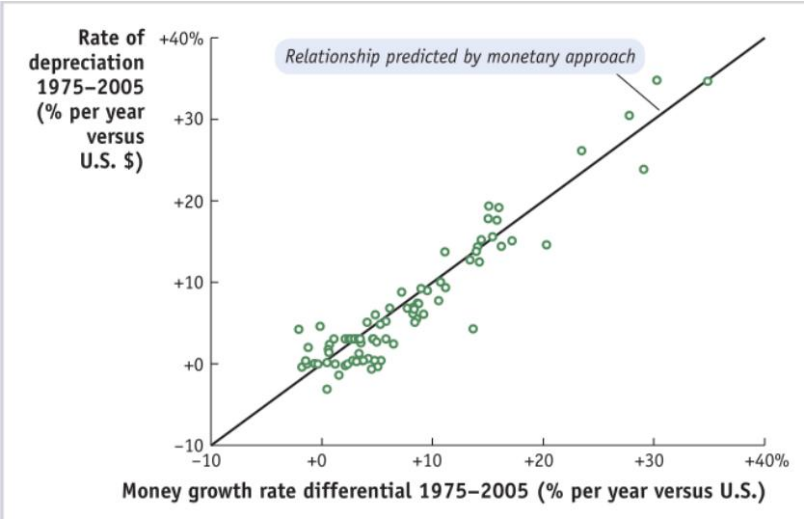
- Combine to get the real interest rate parity

$$r - r^* = \frac{q^e - q}{q}$$

A General Theory of FX in the Long Run

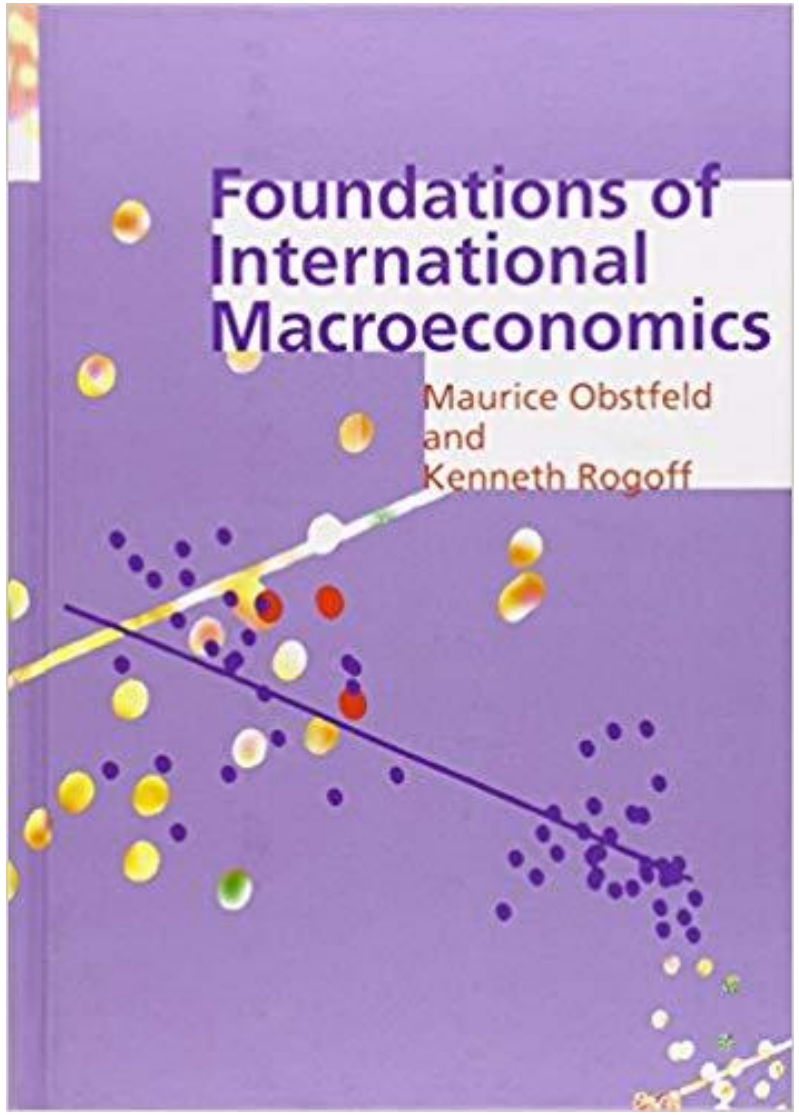
Demand, Supply, and Real FX

- Small deviations from 45-degree line can be thought of as variations in the long-run real exchange rate



- We postulate that the long-run values of real exchange rates depend on relative demand and supply conditions.

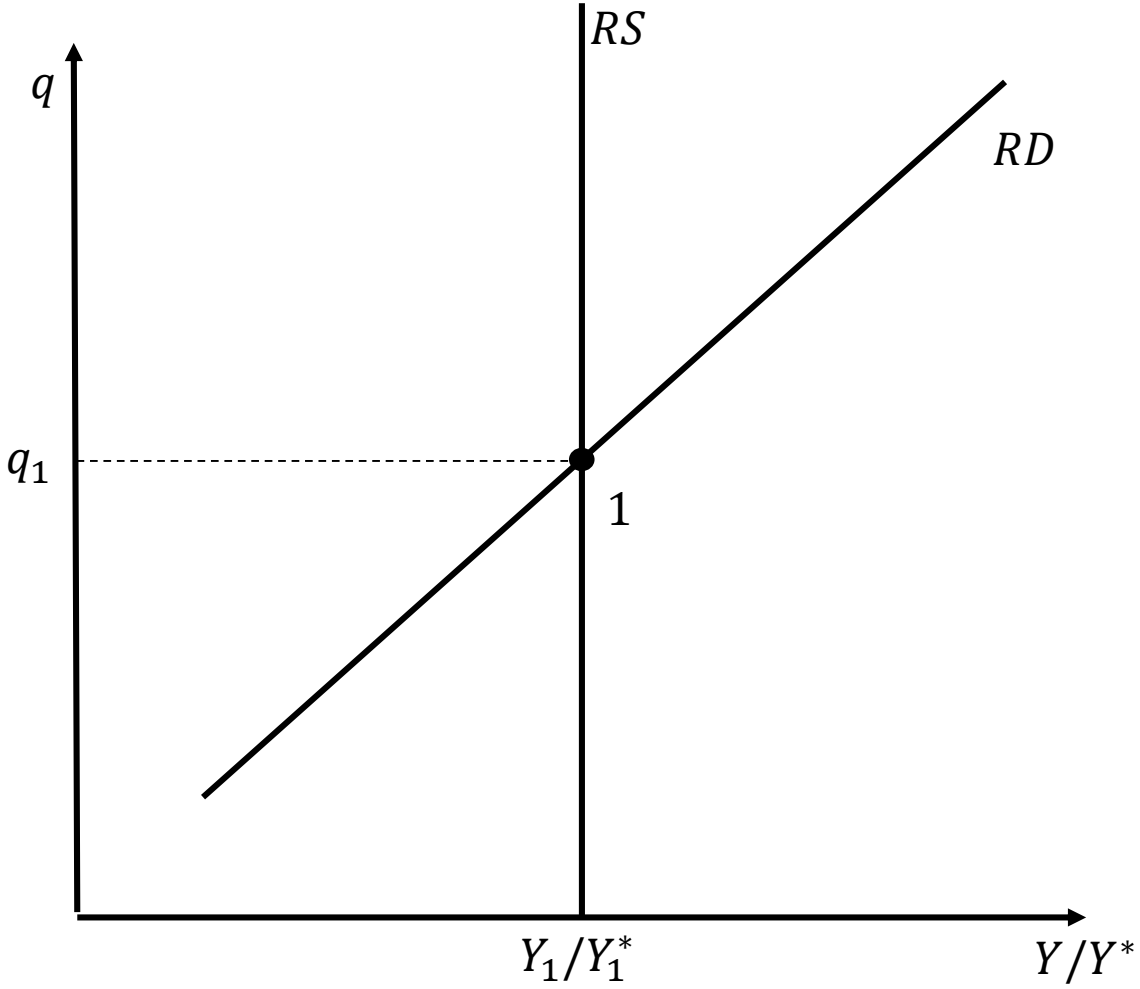
Demand, Supply, and Real Exchange Rate



- A classic in-depth discussion of the real exchange rate determination is Chapter 4 of Obstfeld and Rogoff (1996)

- We will simplify everything using a straightforward KOM book diagram!

Relative Demand and Relative Supply



Relative Demand and Relative Supply

Intuition

- Relative demand for domestic products:
  - If the relative price of domestic goods decreases ($q \uparrow$), demand for domestic goods relative to foreign goods increases: *RD is upward sloping*

- Relative supply of domestic products:
  - In LR, output is determined by supply factors and not affected by q: *RS is a vertical line*

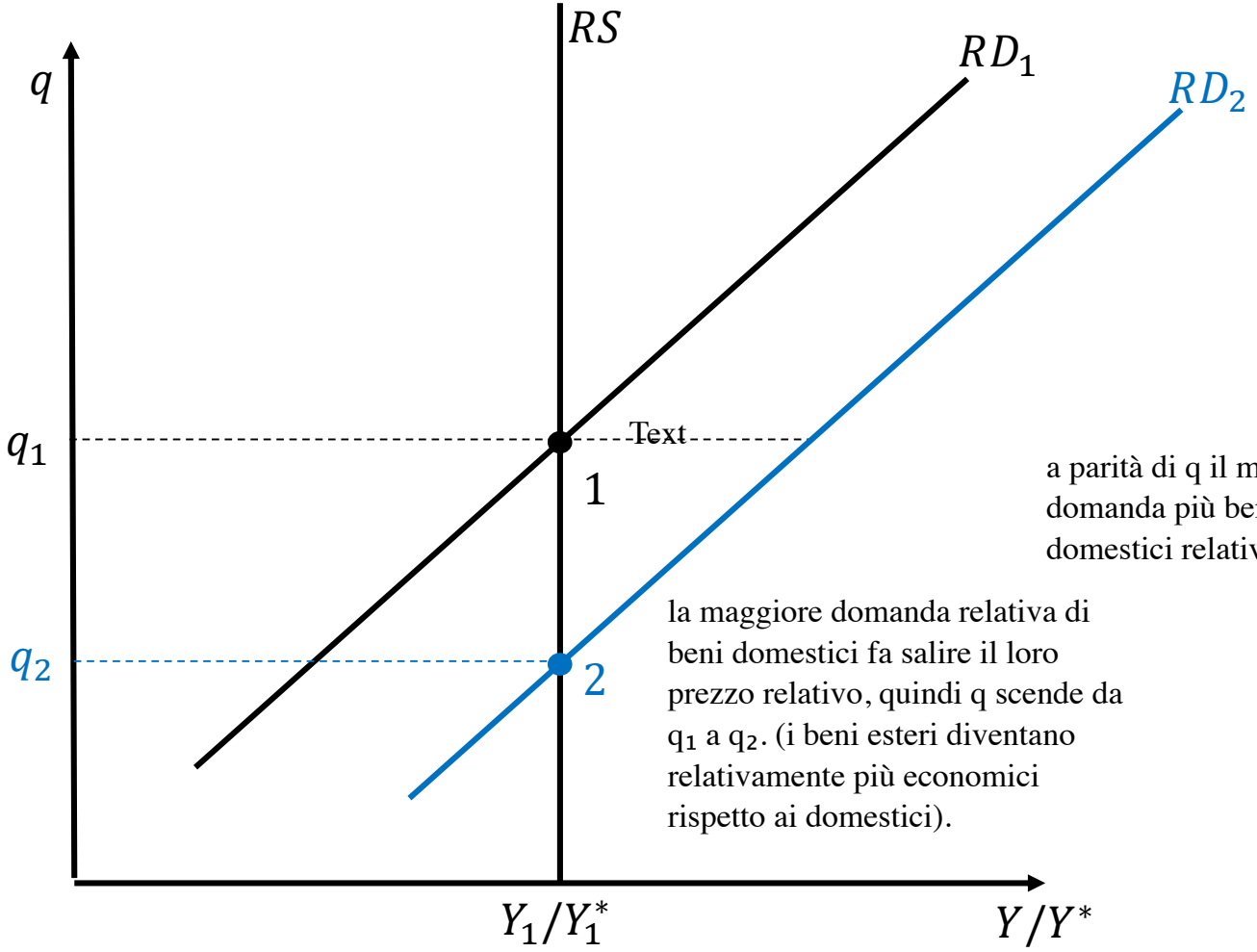
Exercise 1

A change in relative demand

Q è domestic per foreign, quante unità di paniere estero compri con un paniere domestico

q = E·(P_E / P_US)

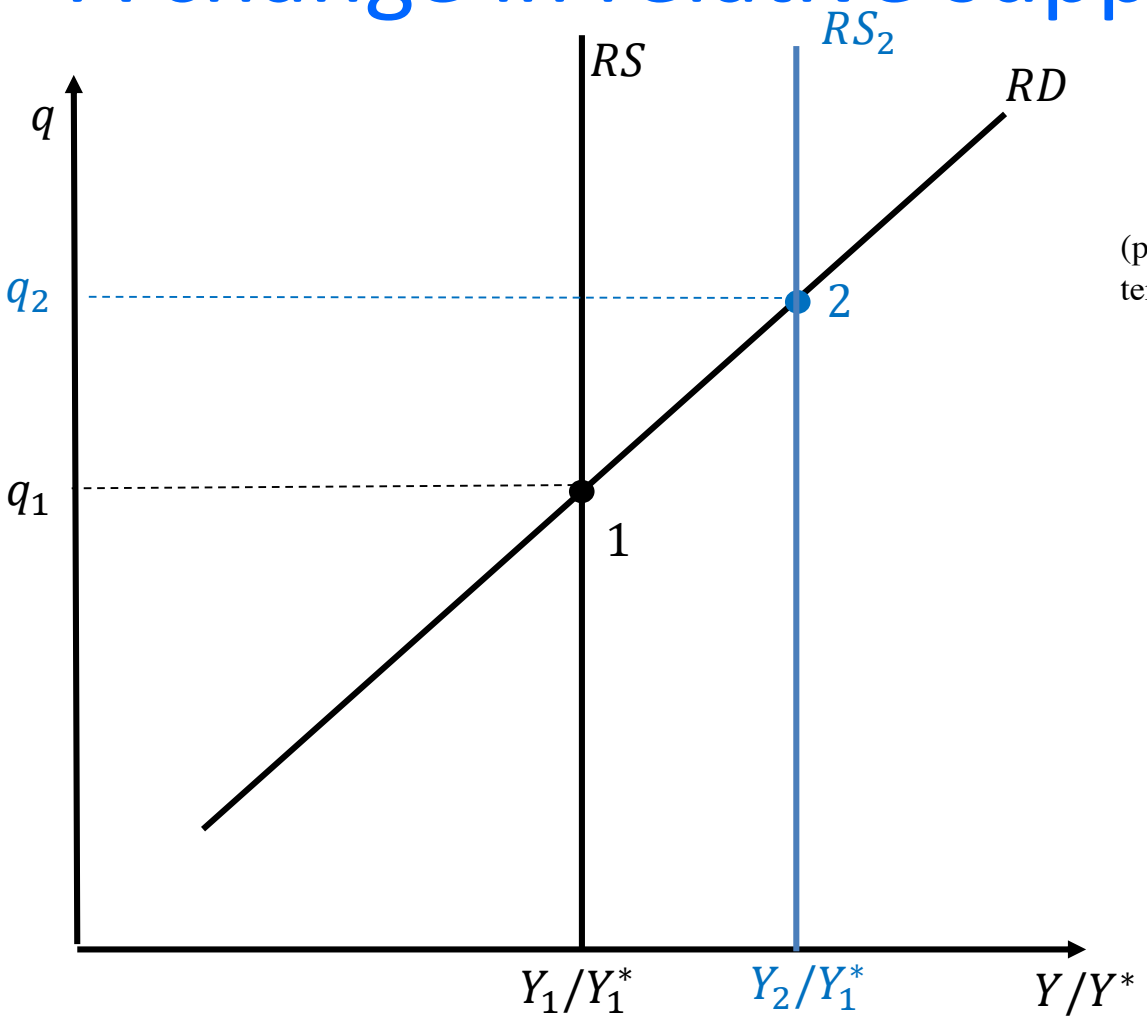
(prezzo relativo dei beni domestici in termini di beni esteri)



a parità di q il mondo domanda più beni domestici relativi

la maggiore domanda relativa di beni domestici fa salire il loro prezzo relativo, quindi q scende da q_1 a q_2. (i beni esteri diventano relativamente più economici rispetto ai domestici).

Exercise 2
A change in relative supply



(prezzo relativo dei beni domestici in termini di beni esteri)

A General Theory of Long-Run Exchange Rates

- Definition of real ER

$$q_{\$/€} = \frac{E_{\$/€} \cdot P_E}{P_{US}}$$

- A general theory of nominal ER

$$E_{\$/€} = q_{\$/€} \frac{P_{US}}{P_E}$$

A General Theory of Long-Run Exchange Rates

$$E_{\$/€} = q_{\$/€} \frac{P_{US}}{P_E}$$

- If only monetary factors change, then same predictions as before
  - No changes in the real exchange rate occurs
- When factors influencing real output change, the real exchange rate changes (and possibly price levels)

Exercise

An increase in the relative supply of domestic products

- The effect on real exchange rate

$$RS \uparrow \Rightarrow q \uparrow \Rightarrow E \uparrow$$

- The effect on price levels

$$P = \frac{M^S}{L(Y,R)}$$

  - $RS \uparrow \Rightarrow Y \uparrow \Rightarrow L(Y,R) \uparrow \Rightarrow \ P \downarrow \Rightarrow E \downarrow$

- Ambiguous effect on the nominal exchange rate

$$E = q \uparrow \frac{P \downarrow}{P^*}$$

From Fixed to Variable Output

Previous Lectures

- Assumption so far: short-run output is exogenous (and fixed)

- But all changes that we studied can potentially affect output in the short run

- The next several lectures
  - **Endogenous** output in the short run

Output

- The long run
  - Prices are flexible
  - Output is at its full employment level (aka, natural level of output) determined by supply factors: labor, capital, technology
  - "Aggregate demand" does not matter

- The short run
  - Prices are sticky
  - Aggregate demand affects output
    - Why? Imagine what happens when you decide to consume one more pizza when its price is fixed

Aggregate Demand

- Aggregate demand
  - The amount of a country's goods and services demanded by households and firms around the world (Y)
- Recall the identity $Y = C + I + G + NX$
- The aggregate demand for an open economy's output consists of four components
  - Consumption demand (C)
  - Investment demand (I): <u>assumed fixed</u>
  - Government demand (G): <u>assumed fixed</u>
  - Net exports (NX)

Consumption

Consumption Demand

- Desired consumption of *household i*

$$C_i = C(Y_i^d; r, Y_i^{d,e}, W_i)$$

  - Y_i^d - individual *disposable* (**it means after-tax income here**) income
  - $Y_i^{d,e}$ - expected future individual disposable income
  - r - **real** interest rate
  - W_i - wealth
- Aggregate desired consumption

$$C = C(Y^d; r, Y^{d,e}, W)$$

  - A sum of desired consumption levels is a function of a sum of disposable incomes
  - Assumption (consumption smoothing): $\left|\frac{dC}{dY^d}\right| < 1$

Net Exports

Net Exports

- We assume that NX is primarily determined by two factors

$$NX = NX\left(\frac{EP^*}{P}, Y^d\right)$$

  – Other factors can also be important, but we simplify our lives

- Note: the book uses CA and calls it net demand for goods and services!
  – I like to use NX because we can talk about the demand for goods and services produced at home (GDP)
  – To make my slides and the book consistent, you can assume that throughout this analysis NIFA=0

When does NX depend on q?

- Consider trade between Italy (home country) and the UK (foreign country)
- Italian net exports are

$$NX = Ex - Im$$

  – *NX, Ex, Im* are **measured in units of Italian (domestic) output**

- Trade flows and prices

$$Ex = Ex\left(\frac{P^*}{P^{\text{Im from ITA in £}}}\right) = Ex\left(\frac{P^*}{P/E}\right) = Ex\left(\frac{P^*E}{P}\right) = Ex(q)$$

$$Im = \frac{P^{\text{Im from UK in €}}}{P} Ex^*\left(\frac{P}{P^{\text{Im from UK in €}}}\right) = \frac{P^*E}{P} Ex^*\left(\frac{P}{P^*E}\right) = q \cdot Ex^*\left(\frac{1}{q}\right)$$

  – where I assumed that the prices of internationally traded goods are *set (and sticky)* in producers currency [**producer-currency pricing or PCP**]

How do changes in *q* affect *NX*?

$$NX(q) = Ex(q) - q \cdot Ex^*\left(\frac{1}{q}\right)$$

- Where I omitted the explicit dependence on output for simplicity

- Exports
  - An increase in domestic real ER q (i.e., a real depreciation of domestic currency) increases foreign demand for domestic goods: $\frac{dEx}{dq} > 0$
- Imports
  - How does an increase in q affect *Im*?

How do changes in *q* affect *Im*?

- *Im* denotes the value of imports measured in terms of domestic output:

$$Im = q \cdot Ex^* \left(\frac{1}{q}\right)$$

- Take a derivative

$$\frac{dIm}{dq} = \underbrace{1 \cdot Ex^*}_{\substack{\text{Value effect} \\ >0}} + \underbrace{q\frac{dEx^*}{dq}}_{\substack{\text{Volume effect} \\ <0}}$$

How do changes in *q* affect *NX*?

- Combine $\frac{dEx}{dq}$ *and* $\frac{dIm}{dq}$ *to get* $\frac{dNX}{dq}$

$$\frac{dNX}{dq} = \underbrace{\left(\frac{dEx}{dq} - q\frac{dEx^*}{dq}\right)}_{>0}\underbrace{-Ex^*}_{<0}$$

- Whether NX improves or worsens depends on which effect of a real exchange rate change is dominant
- Two important insights
  - The Marshall-Lerner condition
  - The J-curve